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Effect of hydroponic nutrient composition and the barley variety on saponarin production under white LED

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Abstract

This study investigates the effect of nutrient composition and barley cultivars on saponarin production in young barley leaves (YBL) under low-intensity white LED light. Various treatments involving K_2HPO_4 , NaCl, and salicylic acid (SA) were evaluated to be under 6500 K light with 210 PPFD. Among different nutrient combinations, 600 mg L⁻¹ K_2HPO_4 with 0.1% NaCl significantly enhanced saponarin concentration (12.3 g kg_{dw}⁻¹) and total saponarin amount per plant (150 µg plant⁻¹). In addition, the effects of light intensity on saponarin production were also examined across six barley cultivars, and the high light intensity (500 PPFD) notably increased total saponarin amount (212 µg plant⁻¹) in Keunal-bori than Hyenuri cultivars. Otherwise, Hyenuri cultivar showed the highest total saponarin amount (114 µg plant⁻¹) at low light intensity (100 PPFD), as well as the highest electricity efficiency (201 mg kWh⁻¹). These findings provide valuable insights into optimizing cultivation conditions for each cultivar in plant factories to maximize saponarin production while balancing energy consumption and metabolic efficiency.

Keywords Dipotassium phosphate, Barley cultivar, Energy efficiency, Saponarin, Lutonarin, Flavonoid-3'-monooxygenase

Introduction

Photosynthesis and carbon metabolism are greatly influenced by the light quality; in particular, red light (600–700 nm) and blue light (400–500 nm) are essential for these processes in plants [1–4]. Furthermore, the importance of full-spectrum light has recently been recognized due to its beneficial effects on metabolite production and plant growth [5–10]. However, when growing crops

under artificial lighting conditions rather than sunlight, light energy efficiency should be considered. Therefore, precise controls of light quality and nutrient supplementation are a key strategy for optimizing plant growth and enhancing the biosynthesis of carbon-based metabolites in a plant factory under artificial light [11–17]. Recently, plant factories have been increasingly used to produce high-value phytochemicals, such as flavonoids, anthocyanins, and polyphenols. In particular, exposure to blue light and ultraviolet (UV) radiation was demonstrated to stimulate the biosynthesis of antioxidative secondary metabolites in plants [18–20]. However, excessive exposure to short-wavelength (blue and UV) or high-intensity lights could elevate intracellular reactive oxygen species (ROS) levels in plant cells, thereby reducing photosynthetic efficiency and ultimately suppressing plant growth

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[21, 22]. Therefore, optimization studies of the cultivation environment are necessary for each crop and specific target metabolites.

Young barley leaves (YBL) are known to biosynthesize saponarin, a biologically active flavonoid glycoside, including anti-inflammatory and hepatoprotective effects, and it is richly present in YBL [23–25]. In open-field cultivation, the saponarin content was reported to be 11 g kg^{−1} of dry weight [26], and similar saponarin levels were observed in plant factory cultivation under blue light or ultraviolet light treatment conditions; however, the growth suppression was observed in YBL under the blue or UV light conditions, and the biomass production was not good in comparison with the open-field cultivation [27]. To determine the optimal light condition for saponarin production in a plant factory, Oh et al. (2021) investigated the changes in functional flavonoid content in response to correlated color temperature and cycle through a study on artificial light composition for saponarin biosynthesis in YBL [28]. Similarly, Lee et al. (2024) reported that saponarin biosynthesis flux in YBL was controlled by light-induced ROS [14]. Finally, they achieved saponarin production levels comparable to those observed in open-field cultivation under high-light intensity (≥ 500 PPFD). However, the proposed high-light intensity treatments were impractical for commercial application in plant factories due to excessive heat generation and elevated energy consumption. Therefore, practical conditions for reducing energy consumption during YBL cultivation need to be studied by optimizing nutrient composition, ROS-related hormones, saponarin precursor, and cultivar effects under low light intensity. Furthermore, comparative targeted gene-expression and metabolite analyses were conducted on the selected cultivars to elucidate the key biosynthetic steps involved in saponarin accumulation in YBL.

Materials and methods

Chemicals and reagents

Standard chemicals for isovitexin, *p*-coumaric acid, saponarin, abscisic acid (ABA), cinnamic acid, sodium salicylate, and phenylalanine were purchased from

Sigma-Aldrich (Merck KGaA, Darmstadt, Germany). Lutonarin was obtained from the National Institute of Crop Science and Food Science (Wanju, Republic of Korea). Calcium chloride dihydrate, magnesium chloride dihydrate, sodium nitrate, ammonium chloride, sodium phosphate monobasic, potassium chloride, and potassium phosphate dibasic were used for nutrient minerals with Daejung Chemicals & Metals Co., Ltd. (Siheung, Republic of Korea). Water and methanol were used HPLC grade from Burdick & Jackson™ (Honeywell International Inc., Morris Plains, NJ, USA).

YBL cultivation

The barley (*Hordeum vulgare* L.) seeds for cultivation were supported by the National Institute of Crop Science and Food Science (Wanju, Republic of Korea). The hydroponic cultivation conditions described in the previous report were followed [14]. Briefly, 50 g of seed was soaked in tap water for two days, then the sprouted seed was transferred to a tray (0.21 × 0.27 m) and cultivated in hydroponic conditions under white LED lights (3000 K and 6500 K lights, Kkumi Lighting Co., Ltd., Namyangju, Republic of Korea) at 20–22°C. Light intensity was adjusted to 100, 210, and 500 μmol m^{−2} s^{−1} (PPFD) and measured with a Portable Luxmeter (HD2102.1, Delta OHM S.r.l., Selvazzano Dentro, Italy). The light cycle was 12 h a day, and the primary leaf was harvested at 15 cm. Each hydroponic cultivation was replicated with the three separate trays under each cultivation condition. The harvested leaf was dried at 45°C in an air-convection dryer (PS-1000 A, Shinil Tech Co., Ltd., Gimhae, Republic of Korea), then ground and stored at −20°C until instrumental analysis.

Analysis of saponarin and metabolites

Saponarin and the metabolites, including 4-coumaric acid, isovitexin, and lutonarin, were analyzed by the previously reported method [14]. Briefly, 1.0 g of dried YBL was added to 30 mL of 60% MeOH and shaken for 4 h at 4000 rpm. The extract was centrifuged for 10 min, and the supernatant was filtered with a 0.45 μm PVDF syringe filter (Biofact Co., Ltd., Daejeon, Republic of Korea). The filtrate was analyzed with HPLC-UV (Agilent 1200, Agilent Technologies, Inc., Santa Clara, CA, USA). The detailed instrumental condition was described in Table 1. Each tray sample was analyzed in triplicate, and the average was presented.

Electricity energy use for saponarin production

25 W of 6500 K White LED used for the cultivation of YBL, and the LEDs were installed at 0.15, 0.31, and 0.62 kW m^{−2} to radiate 100, 210, and 500 PPFD for nine cultivation period in the cultivation chamber. To evaluate saponarin production of energy efficiency, 850 g m^{−2}

Table 1 HPLC instrumental conditions for saponarin

Instrument	Agilent 1200 LC system		
Column	C18 (3.0 × 100 mm, 3.5 μm)		
Injection volume	2 μL		
Flow rate	1 mL min ^{−1}		
Detection	325 nm		
Mobile phase	A: 0.1% formic acid in dH ₂ O B: acetonitrile		
Gradient	Time (min)	A (%)	B (%)
	0	95	5
	21	0	100

of barley seeds were used for the YBL cultivation under YBL. The thousand-seed weights for Keunal-bori and Hyenuri were used 38.3 g and 29.8 g, respectively. Energy efficiency for saponarin production (EESP) was calculated using Eq. 1. Finally, saponarin production efficiency per energy consumption (mg kWh^{-1}) was calculated.

$$EESP \text{ (mg kWh}^{-1}\text{)} = \frac{\text{Total saponarin production (mg m}^{-2}\text{)}}{\text{Total electric energy use (kWh m}^{-2}\text{)}} \quad (1)$$

RNA expression analysis for CHI, CHS, OGT, and F3'M

Total RNA was extracted from YBL in different treatment groups using IQeasy Plus Plant RNA Extraction Kit (iNtRON Biotechnology, Seongnam, Republic of Korea) according to the manufacturer’s instruction. The quantity and quality of the RNA was determined using a BioPhotometer D30 (Eppendorf, Hamburg, Germany). Total RNA concentration was then adjusted to 50–100 ng μL^{-1} before stored at -80°C immediately after extraction. Extracted RNA was reverse-transcribed into complementary DNA (cDNA) using an M-MLV cDNA Synthesis Kit (Enzynomics, Daejeon, Republic of Korea) according to the manufacturer’s protocol. RNA expression level was determined by quantitative real-time polymerase chain reaction (qRT-PCR, QuantStudio 1, Applied Biosystems, MA, USA) with a TOPreal™ SYBR Green qPCR PreMIX (Enzynomics, Daejeon, Republic of Korea). GAPDH1 was used as a reference gene for quantitative gene expression analysis in YBL. The primers for qRT-PCR analysis are listed in Table 2. YBL for quantitative RNA expression analysis were sampled from three separate batches, and the expression data were presented as relative average values with error bars, normalized to qGAP1 expression. All the targeted gene expression data were presented in comparison with the non-nutrient treatment batch.

Statistical analysis

All cultivation experiments and instrumental analyses were performed with three replications, and statistical data were obtained using the R statistical program (version 4.2.3). Differences in treatment groups were

explained using two-way ANOVA, and Duncan’s multiple range test (DMRT) was used for analysis when significance was confirmed. For all analyses, the significance level was set at $p < 0.05$.

Results and discussion

Effect of nutrient treatments and light conditions on saponarin production in YBL

The effects of major nutrient supply (N, P, K, Ca, and Mg) on saponarin and biomass production in YBL were investigated under two different 6500 K and 3000 K at $210 \mu\text{mol m}^{-2} \text{ s}^{-1}$. At 6500 K, the P/K mixture treatment resulted in the highest saponarin concentration, reaching $7.82 \text{ g kg}_{\text{dw}}^{-1}$ (Table 3). This was followed closely by the PO_4^{3-} and N/P/K (pH 6) treatments, which recorded concentrations of $7.29 \text{ g kg}_{\text{dw}}^{-1}$ and $7.31 \text{ g kg}_{\text{dw}}^{-1}$, respectively. On the other hand, all treatments at 3000 K showed lower saponarin concentrations than those at 6500 K; the P/K treatment showed the highest concentration at $6.13 \text{ g kg}_{\text{dw}}^{-1}$, although this was notably lower than the concentration observed at 6500 K. In addition, biomass production was observed to have no significant effect from the treatments. This demonstrated blue-rich white light (6500 K) had a key factor in saponarin biosynthesis in YBL, and it was a similar result to previous reports [14]. Finally, under 6500 K light, the P/K treatment achieved the highest total saponarin amount at $79.0 \mu\text{g plant}^{-1}$, followed by the PO_4^{3-} ($70.7 \mu\text{g plant}^{-1}$) and the N/P/K (pH 6) ($70.2 \mu\text{g plant}^{-1}$) treatments, while the P/K treatment under 3000 K light showed the highest total saponarin amount ($57.6 \mu\text{g plant}^{-1}$). Thus, the P/K treatment consistently resulted in the highest saponarin concentration and total saponarin amount in both light conditions, and 6500 K light treatment was identified as more conducive to saponarin production in YBL. However, these values were not as high as those observed in YBL grown under high light intensity or in an open field condition.

Effect of plant hormones on saponarin in YBL

Two plant hormones and three precursors of saponarin biosynthesis were investigated for total saponarin amount in YBL grown under 6500 K light with an irradiance of 210 PPFD. Salicylic acid (SA) was shown to be a potent enhancer of saponarin production. The treatment with 10.0 mg L^{-1} SA resulted in the highest saponarin concentration of $7.5 \text{ g kg}_{\text{dw}}^{-1}$ and a biomass of $9.9 \text{ mg}_{\text{dw}} \text{ plant}^{-1}$, leading to a substantial total saponarin amount of $74.3 \mu\text{g plant}^{-1}$ (Table 4). Interestingly, the total saponarin amount was highest at a lower concentration of 1.0 mg L^{-1} SA, reaching $75.2 \mu\text{g plant}^{-1}$, but this difference was not statistically significant. This suggests that SA treatment could enhance saponarin concentration in YBL; however, a high concentration (over 50 mg

Table 2 Primers sequences for qRT-PCR analysis

Gene	Forward (5' → 3')	Reverse (5' → 3')
HvCHS1	TTTCATCATGGACGAGATGC	TGTGGAGGACGACGGTCT
HvCHI	AACTGTGTCCGCTACTGGAA	GGGGAGTGCCTGAAGAGTAT
HvOGT1	TGTCGAGATAGTTTGTCCGGTGTG	ATCACGCCGTCCGATG-GATATCTG
HvOGT2	CAGCAGGAGGAACCTGGATA	CTGGCACCAAGCAGCAGAT
HvF3'M1	CATATGCTATGCGTCTTTC	ACATCTCGTGCAATGTCT
HvF3'M2	GGGAATCCGGAAGAAGTC	TTGAGCATGTATTTCGAGGAA
HvF3'M3	GATGGACGGATTATCAGC	GTGTCAATTGAAGTTATCCC
qGAP1	AGAAGGAATACAAGTCAGACATCG	CTCTCCAGTCCTTGCTTGAA

Table 3 Nutrients and pH effect on the mean saponarin amount in dried barley sprout (Keunal-bori cultivar) under 6500 K and 3000 K color temperature of light with $210 \mu\text{mol m}^{-2} \text{s}^{-1}$

Treatment		6500 K			3000 K		
Nutrient	Concentration (mg L^{-1})	Saponarin ($\text{g kg}_{\text{dw}}^{-1}$)	Biomass ($\text{mg}_{\text{dw}} \text{plant}^{-1}$)	Total saponarin ($\mu\text{g plant}^{-1}$)	Saponarin ($\text{g kg}_{\text{dw}}^{-1}$)	Biomass ($\text{mg}_{\text{dw}} \text{plant}^{-1}$)	Total saponarin ($\mu\text{g plant}^{-1}$)
Non-treatment	-	6.51b	10.3a	67.1b	4.64ab	10.0a	46.4b
NO_3^-	100	6.46b	9.8a	63.3ab	4.67ab	8.8b	41.1ab
NH_4^+	100	6.21ab	9.9a	61.5ab	4.45ab	8.9ab	39.6a
PO_4^{3-}	100	7.29c	9.7a	70.7b	4.20a	9.3ab	39.1a
K^+	100	6.90bc	10.0a	69.0b	5.05b	9.4ab	47.5b
Ca^{2+}	100	5.69a	10.0a	56.9a	4.73b	9.3ab	44.0ab
Mg^{2+}	100	5.59a	10.3a	57.6a	4.59ab	9.0ab	41.3ab
Ca/Mg	200	6.05ab	10.4a	62.9ab	5.01b	9.4ab	47.1b
N/P	100/100	7.18c	10.4a	74.7bc	5.21b	9.4ab	49.0b
P/K	100/100	7.82d	10.1a	79.0c	6.13c	9.4ab	57.6c
N/P/K ^{a)} (pH 4)	100/50/50	6.23ab	9.8a	61.1ab	4.66ab	9.5ab	44.3ab
N/P/K (pH 6)	100/50/50	7.31c	9.6a	70.2b	4.93b	9.6a	47.3b
N/P/K (pH 8)	100/50/50	6.64bc	9.8a	65.1b	5.41bc	8.9ab	48.1b

^{a)} N, P, and K were supplied as NaNO_3 , NaH_2PO_4 , and KCl, respectively

Table 4 Effect of the plant hormone treatment on the saponarin content in the YBL under 6500 K light (210PPFD)

	Treatment	Concentration (mg L^{-1})	Saponarin ($\text{g kg}_{\text{dw}}^{-1}$)	Biomass ($\text{mg}_{\text{dw}} \text{plant}^{-1}$)	Total saponarin ($\mu\text{g plant}^{-1}$)
Hormone	NT ^{a)}	-	6.5a	10.3a	66.8ab
	SA	0.2	6.2a	11.3b	71.1b
		1.0	6.9ab	10.9ab	75.2b
		5.0	7.3b	10.1a	73.7b
		10	7.5b	9.9a	74.3b
		50	6.2a	9.6a	63.5ab
	ABA	0.005	6.0a	9.8a	59.2a
		0.02	6.3a	9.4a	61.4a
Precursors	PA	50	6.2a	9.9a	61.4a
		250	6.5a	10.1a	65.6ab
	CNA	50	6.1a	10.1a	61.6a
		250	6.2a	9.9a	61.9a
	CMA	50	6.6a	9.9a	65.2ab
		250	6.5a	9.4a	61.8a

^{a)} NT means non-treatment

L^{-1}) could result in lower saponarin levels and lower biomass. In contrast, ABA demonstrated a less pronounced effect on saponarin accumulation. At 0.005 mg L^{-1} , ABA resulted in a saponarin concentration of $6.0 \text{ g kg}_{\text{dw}}^{-1}$ and a total saponarin amount of $59.2 \mu\text{g plant}^{-1}$, with a biomass of $9.8 \text{ mg}_{\text{dw}} \text{plant}^{-1}$. Increasing the concentration to 0.02 mg L^{-1} led to a slight increase in saponarin concentration ($6.3 \text{ g kg}_{\text{dw}}^{-1}$), but it decreased biomass, resulting in a total saponarin amount of $61.4 \mu\text{g plant}^{-1}$ with no statistically significant. At concentrations above 0.1 mg L^{-1} , YBL growth was significantly suppressed. These findings suggested that while ABA would influence saponarin production, its overall efficacy was lower than that of SA.

Effect of the saponarin precursor supplement on saponarin in YBL

The precursor treatment effect on saponarin production was investigated using phenylalanine (PA), cinnamic acid (CNA), and coumaric acid (CMA) in YBL. However, no effects of precursor supplement up to 250 mg L^{-1} were observed on saponarin accumulation and the biomass increment of YBL. In conclusion, SA, particularly at lower concentrations, stands out as a powerful tool for maximizing saponarin production, whereas ABA and precursor supplements in hydroponics would not positively affect total saponarin increment significantly.

Effect of nutrient combination with K/P, NaCl, and SA on saponarin content in YBL

Based on the K/P supplement effect on saponarin increment with the previous report by Lee et al. [14], the nutrient supplements were designed with K_2HPO_4 and $0.1 \text{ wt\% NaCl}$ in hydroponics. In the no nutrient supplement group as a control under $210 \mu\text{mol m}^{-2} \text{s}^{-1}$, the saponarin concentration was recorded at $6.51 \text{ g kg}_{\text{dw}}^{-1}$, with a biomass of $10.3 \text{ mg}_{\text{dw}} \text{plant}^{-1}$, leading to a total saponarin amount of $67.1 \mu\text{g plant}^{-1}$ (Table 5). The K/P with $0.1 \text{ wt\% NaCl}$ treatments were particularly effective in increasing the saponarin content. At a treatment of $200 \text{ mg L}^{-1} \text{ K}_2\text{HPO}_4$, the saponarin concentration increased to $11.5 \text{ g kg}_{\text{dw}}^{-1}$, with a biomass of $9.3 \text{ mg}_{\text{dw}} \text{plant}^{-1}$, resulting in a total saponarin amount of $107 \mu\text{g plant}^{-1}$. Increasing the K_2HPO_4 concentration to 600 mg L^{-1} further enhanced the saponarin concentration to $12.3 \text{ g kg}_{\text{dw}}^{-1}$, with a biomass of $12.2 \text{ mg}_{\text{dw}} \text{plant}^{-1}$, and a total saponarin amount of $150 \mu\text{g plant}^{-1}$. Although P/K treatments at low concentration (200 mg L^{-1}) didn't affect the biomass increase (in Table 3), this increase was observed at $600 \text{ mg L}^{-1} \text{ K}_2\text{HPO}_4$. These nutrient concentration

Table 5 Nutrient and pH effects on saponarin amount in dried barley sprouts at 6500 K light at 210 $\mu\text{mol m}^{-2} \text{s}^{-1}$

Nutrient	Concentration (mg L^{-1})	Saponarin ($\text{g kg}_{\text{dw}}^{-1}$)	Biomass ($\text{mg}_{\text{dw}} \text{plant}^{-1}$)	Total saponarin ($\mu\text{g plant}^{-1}$)
Control	-	6.51a	10.3b	67.1a
K/P/Na ^{a)}	200 ^{b)}	11.5b	9.3b	107c
K/P/Na	600 ^{b)}	12.3b	12.2c	150e
K/P/Na	1800 ^{b)}	11.4b	10.9b	124d
K/P/Na ^{c)} + SA	0.2 ^{d)}	11.3b	9.6b	109c
K/P/Na ^{c)} + SA	1.0 ^{d)}	10.6b	9.5b	101c
K/P/Na ^{c)} + SA	5.0 ^{d)}	10.9b	7.7a	83.9b

^{a)}K/P/Na was a composite with K_2HPO_4 and 0.1% NaCl; ^{b)}the concentration was presented for K_2HPO_4 ; ^{c)}K/P/Na was a composite with 600 mg L^{-1} K_2HPO_4 and 0.1% NaCl; ^{d)}the concentration was presented at mg L^{-1} for SA

effects on biomass production were consistent with the results reported in the previous study [14]. However, at 1800 mg L^{-1} of K_2HPO_4 , both the saponarin concentration and the biomass slightly decreased, resulting in a total saponarin amount of 124 $\mu\text{g plant}^{-1}$. Thus, the optimum concentration of K_2HPO_4 would be 600 mg L^{-1} with 0.1% NaCl.

The treatments of SA involving K/P/Na also showed a higher total saponarin amount than the control; however, the total saponarins were lower than those in the SA non-treatment. Briefly, at 0.2 mg L^{-1} of SA with K/P/Na treatment, the saponarin concentration was 11.3 $\text{g kg}_{\text{dw}}^{-1}$, with a biomass of 9.6 $\text{mg}_{\text{dw}} \text{plant}^{-1}$, leading to a total saponarin amount of 109 $\mu\text{g plant}^{-1}$. Also, increasing the SA concentration to 1.0 mg L^{-1} resulted in a slightly lower saponarin concentration (10.6 $\text{g kg}_{\text{dw}}^{-1}$), with a biomass of 9.5 $\text{mg}_{\text{dw}} \text{plant}^{-1}$, and a total saponarin amount (101 $\mu\text{g plant}^{-1}$) was significantly lower than in the no SA treatment (150 $\mu\text{g plant}^{-1}$). These decreases were also observed in the 5.0 mg L^{-1} SA treatment; both biomass (7.7 $\text{mg}_{\text{dw}} \text{plant}^{-1}$) and total saponarin amount (83.9 $\mu\text{g plant}^{-1}$) were decreased. Thus, 600 mg L^{-1} of K_2HPO_4

and 0.1% NaCl significantly enhanced the saponarin production in YBL under 210 PPFD of 6500 K light, and it was comparable to saponarin production with open-field cultivation, while the treatment of SA with this suggested nutrient combination negatively affected the total saponarin production. These findings provide a valuable design for optimizing the cultivation conditions to maximize the production of saponarin-rich barley sprouts.

Effect of barley cultivars on saponarin production under different light intensities

Six different barley cultivars, including Keunal-bori, Sakirang, Saknuri, Heimi, Heyang, and Hyenuri, were compared on saponarin production yield under low (100 PPFD) and high (500 PPFD) light intensity to investigate the light intensity effects. Under high intensity, Hyenuri exhibited the highest saponarin concentration at 14.98 $\text{g kg}_{\text{dw}}^{-1}$ (Table 6), but not the total saponarin amount (157 $\mu\text{g plant}^{-1}$). However, Keunal-bori exhibited the highest total saponarin amount (212 $\mu\text{g plant}^{-1}$), primarily due to its relatively large biomass. Notably, Keunal-bori also exhibited the greatest increase ratio in total saponarin increment (84%) by nutrient supplementation, indicating a robust response to the combined effects of high light and added nutrients. Other cultivars, such as Sakirang and Heimi, also showed significant increases in total saponarin amounts with the nutrient supplement under high light conditions, with amounts of 205 $\mu\text{g plant}^{-1}$ (50%) and 189 $\mu\text{g plant}^{-1}$ (39%), respectively. In the low light condition, a reduction in saponarin production was observed across all cultivars compared to the high light, although the nutrient treatment effect was observed in total saponarin increment (39–82%). Under low-light intensity, Hyenuri exhibited the highest saponarin concentration (12.03 $\text{g kg}_{\text{dw}}^{-1}$) and total saponarin amounts (114 $\mu\text{g plant}^{-1}$).

Table 6 Comparison of saponarin amount by barley cultivars under the high and low light intensity of 6500 K light

Light intensity	Cultivar	Saponarin ($\text{g kg}_{\text{dw}}^{-1}$)	Biomass ($\text{mg}_{\text{dw}} \text{plant}^{-1}$)	Total saponarin ($\mu\text{g plant}^{-1}$)	Total saponarin increment (%) ^{a)}
High (500 PPFD)	Keunal-bori	13.60c	15.6d	212d	84%
	Sakirang	13.70c	15.0d	205d	50%
	Saknuri	9.70b	16.7d	162c	64%
	Heimi	13.14c	14.4c	189 cd	39%
	Heyang	12.26c	13.7bc	168c	44%
	Hyenuri	14.98d	10.5ab	157c	38%
Low (100 PPFD)	Keunal-bori	9.32b	11.3ab	105ab	43%
	Sakirang	9.79b	10.9ab	106ab	39%
	Saknuri	7.74a	12.2b	94.5a	50%
	Heimi	10.02bc	9.8a	98.2a	51%
	Heyang	10.09bc	10.5ab	106ab	62%
	Hyenuri	12.03c	9.5a	114b	82%

^{a)} The ratio for increment of total saponarin amount by potassium and sodium salt (600 mg L^{-1} K_2HPO_4 and 0.1% NaCl) supplement in comparison with no nutrient treatment in YBL

Table 7 Electricity energy use for saponarin production in YBL

PPFD	Energy consumption (kWh m ⁻² day ⁻¹)	Saponarin production per electricity use (mg kWh ⁻¹) ^{a)}	
		Keunal-bori	Hyenuri
100	1.8	144	201
210	3.7	99.4	Na ^{b)}
500	7.4	70.3	66.9

^(a) The YBL cultivation was supplied with 600 mg L⁻¹ K₂HPO₄ and 0.1% NaCl at 6500 K light, ^(b) Not available

Table 8 Relative concentration ratios of the saponarin-related metabolites in YBL in the nutrient supplement condition by cultivars and light intensity

Compounds	Variety (Keunal-bori/Hyenuri)		Light intensity (500PPFD/100PPFD)	
	100PPFD	500PPFD	Keunal-bori	Hyenuri
4-Coumaric acid	1.2	0.5	0.3	0.7
Isovitexin	0.5	0.5	0.5	0.4
Lutonarin	1.1	0.8	1.9	2.6
Saponarin	0.7	0.9	1.5	1.2

The saponarin concentration and the biomass were drastically increased under the high-light intensity in Keunal-bori in comparison with the low-light intensity, whereas the changes were relatively low in Hyenuri by light intensity. This result indicates that saponarin metabolism in Keunal-bori could be more sensitive to variations in light intensity than in Hyenuri.

Electricity energy use and saponarin production efficiency

The energy consumptions for 100, 210, and 500 PPFD were calculated to 1.8, 3.7, and 7.4 kWh m⁻² day⁻¹, and the total energy uses were calculated to be 16.2, 33.5, and 67.0 kWh m⁻² until the harvest, respectively. Based on the total saponarin production, the energy efficiencies for saponarin production in YBL were compared in Table 7. Although Hyenuri showed a lower amount of saponarin than Keunal-bori, the energy efficiency for saponarin production of Hyenuri (201 mg kWh⁻¹) showed higher than that of Keunal-bori (144 mg kWh⁻¹) under 100 PPFD. This was likely due to the seedling density of Hyenuri being 1.29 times higher than Keunal-bori. Therefore, Hyenuri could be considered a suitable cultivar for energy-efficient saponarin production in plant factories.

Comparison of metabolites and gene expression levels related to saponarin biosynthesis

To understand the metabolic flow associated with saponarin biosynthesis, the concentrations of its detectable precursors, like 4-coumaric acid and isovitexin, as well as its degradation product, lutonarin, were compared across different light intensities and between the two cultivars (Table 7).

The high-saponarin varieties, Keunal-bori under the high light intensity and Hyenuri under the low light intensity, exhibited higher saponarin and lutonarin concentrations under high light than under low light. The lutonarin concentration in YBL increased by 1.9-fold in Keunal-bori and by 2.6-fold in Hyenuri under the high light intensity compared to low light intensity. However, the lutonarin concentration was not drastically different between the two cultivars in the nutrient supplementation condition (Table 8), whereas the difference showed 2.0-fold higher in Keunal-bori than in Hyenuri in the non-nutrient condition under 500 PPFD (Table S2 and S3). It was presumed that the nutrient supplementation might affect F3'M expression in YBL and the concentrations of saponarin and lutonarin. In addition, the concentrations of precursors for saponarin, like 4-coumaric acid and isovitexin, decreased under the high light intensity in the nutrient supplement condition. Herein, the transcription levels of CHS, CHI, OGT, and F3'M were compared for saponarin accumulation in YBL of two cultivars, Keunal-bori and Hyenuri. CHS, CHI, and OGT were reported to be a potential gene in the saponarin biosynthesis, and F3'M involved in the oxidative degradation of saponarin [29].

After the treatment of K/P/Na under a high (500 PPFD) light intensity, OGT expression was increased at least two-fold, both in the cultivars (Keunal-bori and Hyenuri), while the other saponarin-related gene expression didn't show a significant increment by K/P/Na treatment (Fig. 1). Interestingly, the OGT expression in Hyenuri was more sensitive to the higher light intensity following the K/P/Na treatment than that of Keunal-bori (Fig. 1D); it was 14-fold increased the gene expression under 500 PPFD.

Otherwise, the expression of F3'M was drastically decreased up to 50-fold in Hyenuri by the treatment of K/P/Na under the high light intensity; F3'M1 expression was decreased to 50-fold, and F3'M2 decreased 16-fold. The gene expression in Hyenuri decreased more than in Keunal-bori. This could explain the difference in the accumulation changes of lutonarin, as an oxidation metabolite of saponarin between Keunal-bori and Hyenuri. Thus, Hyenuri exhibited higher F3'M expression reduction and lower lutonarin changes than Keunal-bori by the nutrient treatment under the high light intensity (Fig. 1B and D). These results would explain that the nutrient treatment could lead to a more sensitive decrease in the oxidation of saponarin in YBLs of Hyenuri than in Keunal-bori by suppression of F3'M expression. Thus, the control of F3'M expression in YBL would be a key factor for the saponarin production.

In conclusion, the higher light intensity enhances saponarin production and yield in YBL across all barley cultivars, as well as K/P/Na supplements. In particular,

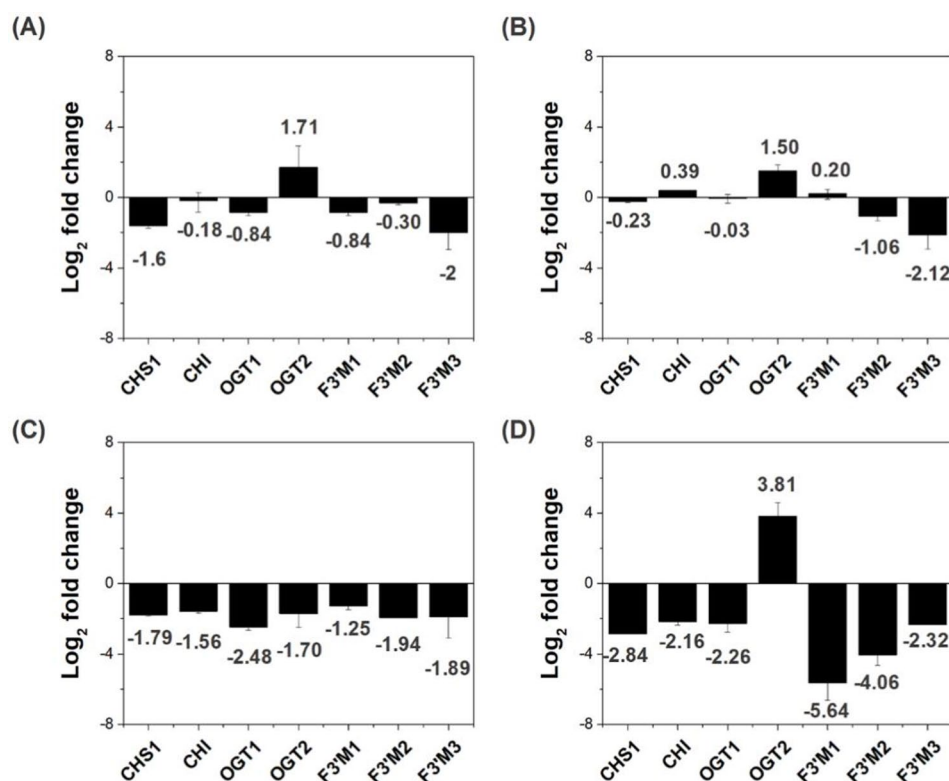


Fig. 1 Relative expression of selected genes in two barley varieties by K/P/Na treatment under different light intensities. Gene expression was analyzed by quantitative qRT-PCR using GAPDH1 as the reference gene. The relative gene expression in Keunal-bori under 100 (A) and 500 PPFD (B) by K/P/Na treatment, and in Hyenuri under 100 (C) and 500 PPFD (D)

Keunal-bori and Hyenuri demonstrated a more sensitive response to saponarin metabolism in YBL by the K/P/Na treatment than the others, resulting in increased saponarin accumulation under both high and low light conditions. Although Keunal-bori showed the highest total saponarin increment in response to the nutrient supplementation under high-light intensity conditions, Hyenuri could be more acceptable for saponarin production under low-energy consumption conditions based on the efficiency of light energy consumption. In particular, the reduced expression of F3'M observed in Hyenuri in response to nutrient treatment could serve as a potential selection marker for enhancing saponarin accumulation and suppressing the oxidation product like lutonarin. However, the effects of F3'M regulation on carbon metabolism, plant growth and stress responses in plant must also be further investigated. Such research will contribute to the development of breeding strategies aimed at selecting barley varieties with high saponarin productivity and stable agronomic traits in plant factories.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13765-026-01134-w>.

Supplementary Material 1.

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Author contributions

Experiment and Instrumental analysis, KY Oh, and SW Kang; data analysis, SI Jung, HJ Seong, KM Kim, YS Jang, and JH Kim; writing and supervision, JH Kim; All authors have read and agreed to the published version of the manuscript.

Data availability

All data supporting the findings of this study are available in this article.

Declarations

Competing interests

The authors declare that they have no competing interests. YS Jang is an Associate Editor of *Applied Biological Chemistry*, yet Associate Editor status has no bearing on the editorial consideration of this article.

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